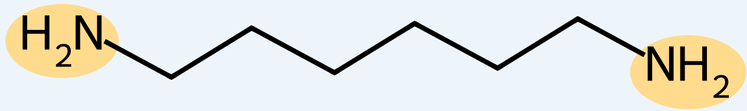
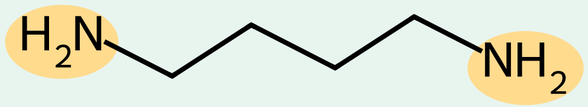
# Native chemistry redirects the enzyme search

**TARGET REACTION**

**Adipic acid + hexamethylenediamine (HMDA)**




Six-carbon chain · two terminal amines

| NATIVE NITROGEN DONOR | | CHEMICAL RELEVANCE | SEARCH DECISION |
|---|---|---|---|
| **GS** | $NH_3$ Ammonia | These native reactions do not establish diamine acceptance. | **Deprioritize** these search scopes |
| **GmaS** | $CH_3NH_2$ Methylamine | | |
| **PuuA** |  Putrescine · four-carbon diamine | A native linear diamine motivates an HMDA lead. | **Expand the search** around the PuuA lead |



**Prioritize by native chemistry; target activity still needs testing.**

Native reactions: IUBMB EC 6.3.1.2 / 6.3.4.12; Kurihara et al., 2008. Actions: archived-run audit, events 51 and 122–125.